Quantitative Research & Trading Framework
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A Structured Literature Review of Quantitative Trading Strategies

Methods, Conditions, and Dependencies

Classical Strategies, Machine Learning, Market Structure, Regimes, Instruments, and Evaluation

Phase 1 Deliverable — Structured Literature Review

A comparative review across ten themes, mapping how strategy, regime, timeframe, instrument, cost, and method interact — concluding with a recommended direction.

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Executive Summary

This review surveys quantitative trading across ten themes and—following the project brief—treats each not as an isolated topic but as one factor in a coupled system. The central organizing question is the brief's own: *under what market conditions, on what instruments, and at what time horizons do which strategies and models become useful?*

Three conclusions emerge. First, no strategy works everywhere: each classical edge (mean reversion, momentum, statistical arbitrage) is contingent on regime, horizon, instrument, and cost structure, and most fail once realistic frictions are imposed. Second, machine learning helps narrowly and conditionally—it is effective for sizing, signal combination, and regime classification, but naive price prediction generally fails because of low signal-to-noise, non-stationarity, and overfitting. Third, honest evaluation is decisive: the Deflated Sharpe Ratio and walk-forward testing are what separate genuine edge from the luckiest of many backtests. The review closes by recommending **PCA-residual statistical arbitrage** (a market-neutral, mid-frequency mean-reversion approach) as the most promising direction for this project, with time-series momentum as a robust alternative.

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1. The Market Setting

Financial markets are noisy, non-stationary, and regime-dependent; predictability, where it exists, is conditional and perishable. Lo's *Adaptive Markets Hypothesis* (2004) provides the organizing lens: market efficiency is not a fixed state but an evolutionary outcome of competition among participants, so opportunities appear where competition is thin and decay as they are exploited. This framing implies that a strategy is a hypothesis about *where and when* structure exists, never a permanent truth.

A strategy is also not merely predictive; per the brief, it must survive transaction costs, slippage, and shifting conditions. This review therefore treats cost realism, microstructure, and regime dependence as first-class constraints rather than afterthoughts. The intended deployment context for the recommended direction is the liquid segment of the Indian equity market at a horizon from intraday to a few weeks, but the survey itself spans horizons from microseconds to months so that the selected direction is chosen against the full landscape.

2. Strategy Families

2.1 Mean reversion

Mean reversion bets that deviations from an equilibrium are temporary. It tends to dominate at short horizons (days), where micro-level supply/demand imbalances correct, and it underpins both pairs trading and the statistical-arbitrage approaches below. Its core assumption—that a divergence is non-fundamental—fails precisely when the divergence is real (genuine news), which is its principal failure mode.

2.2 Momentum and trend following

Momentum bets that recent moves persist. Asness, Moskowitz, and Pedersen (2013) document value and momentum premia across eight diverse markets and asset classes, with a strong common factor structure and a notable negative correlation (around -0.5 to -0.6) between value and momentum. Momentum dominates at medium horizons (one to twelve months) and is especially robust on futures, but it suffers "momentum crashes" at sharp turning points, when recent losers rebound violently against short positions.

2.3 Statistical arbitrage

Statistical arbitrage generalizes relative-value mean reversion from hand-matched pairs to the whole universe. Gatev, Goetzmann, and Rouwenhorst (2006) provide the canonical empirical study of pairs trading, reporting average annualized excess returns of up to roughly 11% on US daily data (1962–2002) that exceed conservative cost estimates, and attribute the profit to temporary mis-pricing of close substitutes linked to a common factor. Avellaneda and Lee (2010) formalize the modern version: remove common factors by PCA, model the residual as a mean-reverting (Ornstein–Uhlenbeck) process, and trade its standardized deviation. Crucially, their reported edge is modest and decaying—PCA-based strategies averaged a Sharpe ratio near 0.9 over 2003–2007—which calibrates realistic expectations.

2.4 Factor strategies

Factor strategies sort the cross-section by characteristics. Fama and French (2015) extend their three-factor model to five factors—market, size (SMB), value (HML), profitability (RMW), and investment (CMA)—and find profitability and investment add explanatory power, though HML becomes partly redundant. Factor strategies operate at monthly horizons with very high breadth, and their performance is itself regime-dependent.

The horizon rule. Short horizons (days) favour reversion; medium horizons (months) favour momentum; factor premia accrue over months with breadth. The same instrument can host opposite strategies at different timescales.

3. Indicators, Patterns, and Microstructure Signals

3.1 Pattern stability

The brief's key question for pattern detection is whether patterns are *statistically significant and stable across periods*. Many candlestick and chart patterns fail this test out of sample. The more durable signals are microstructural and short-lived.

3.2 Order-book imbalance

Among microstructure signals, order-book imbalance—the relative weight of bid versus ask depth—has predictive content for the next price move at the scale of seconds, decaying within minutes. Cartea, Jaimungal, and Penalva (2015) develop the mathematical treatment of such signals and of optimal execution and market making. The practical test is threefold: a signal must be statistically significant, stable across periods, and large enough to survive transaction costs.

3.3 Sentiment signals

Sentiment derived from news or social media is a weak, noisy, but occasionally additive signal at daily horizons. Modern pipelines convert text to scores using finance-specific language models (for example, FinBERT). The hard constraint is data: aligned historical news with timestamps is difficult to source cleanly, which is why sentiment is best treated as an exploratory add-on rather than a core signal.

4. Market Structure

4.1 Microstructure: why spreads exist

Glosten and Milgrom (1985) show that the bid–ask spread arises from adverse selection: a market maker facing possibly-informed traders must quote a spread to avoid systematic losses,

even when risk-neutral and earning zero expected profit. The spread is thus an information cost, set by competition among liquidity providers, and it widens with the proportion of informed flow. Kyle (1985) complements this with a model of how informed trading moves prices, giving the notion of price impact (the linear-in-quantity "Kyle's lambda," often closer to square-root empirically).

4.2 Execution and inventory

Optimal-execution theory (the Almgren–Chriss framework, and the market-making model of Avellaneda and Stoikov) describes how to trade large orders while balancing market impact against timing risk, and how a market maker skews quotes to manage inventory. The unifying microstructure lesson for strategy design is a filter: a signal is only tradable if its expected magnitude exceeds the round-trip cost of acting on it.

4.3 Macro-structure and crisis correlations

At the macro level, interest-rate shifts and geopolitical events change the regime in which a strategy operates. A recurring empirical fact is that correlations spike toward one during crises: assets that normally diversify move together, which is precisely when relative-value and market-neutral strategies are most exposed.

5. Machine Learning and Deep Learning

5.1 Why ML is hard in markets

López de Prado's *Advances in Financial Machine Learning* (2018) catalogues why naive ML fails on financial data: low signal-to-noise (a classifier at 51% accuracy can be excellent), non-stationarity, serially correlated and overlapping labels that leak information, and multiple-testing inflation. The remedies are specific: information-driven bars instead of time bars, fractional differentiation to balance stationarity against memory, triple-barrier labelling, and purged k-fold cross-validation with embargoes to prevent leakage.

5.2 Where ML genuinely helps

The disciplined uses are narrow. *Meta-labelling* separates direction (a primary rule) from sizing (a secondary model that predicts the probability a signal succeeds), and is a robust, proven addition. *Signal combination* via gradient boosting weights multiple trusted signals by condition. *Regime classification* infers the current regime from many features. In each case the model refines decisions on top of a working strategy rather than predicting raw prices.

5.3 Model classes

For tabular features, gradient-boosted trees are the default and frequently the best choice. Sequence models (LSTMs, Transformers) are appropriate only where genuine temporal structure exists and data is abundant; on small, noisy price series they tend to memorize noise. Reinforcement learning for trading from scratch is sample-inefficient and overfits badly, and is best treated as exploratory.

ML discipline. Apply ML to sizing, signal combination, and regime detection—never to direction (which is price prediction) or to hard risk limits (which must stay deterministic). Complexity is added only after a simpler baseline is measured.

6. Strategy Formulation

A predictive signal is not a strategy; the trading rule around it determines viability. The standard mappings are: classification output to long/short/flat; regression output to position size; ranking output to long-top-k/short-bottom-k; regime output to a strategy switch; volatility forecast to risk adjustment; anomaly score to a trade filter; and an ensemble to a meta-strategy. Position sizing is typically volatility-scaled (equal risk rather than equal capital). The Fundamental Law of Active Management, IR $\approx$ IC $\times \sqrt{\text{Breadth}}$, formalizes why many weakly-correlated bets can produce a strong information ratio—provided the bets are genuinely independent, since correlation destroys effective breadth. Risk management (stops, caps, drawdown breakers) is what converts a fragile edge into a survivable one.

7. Stock Selection

Methods for choosing what to trade range from factor sorts (Section 2.4) to clustering algorithms that group similar assets, to price/volume screens, fundamentals, and sentiment. For the relative-value approaches central to this review, selection is implicit: PCA uses the entire liquid universe to estimate common factors, and the trading signal (the standardized residual) selects which names to trade on any given day. Liquidity filtering is the binding practical constraint—illiquid names carry wide spreads and high impact that erode any edge.

8. Arbitrage, Market Shifts, and Regime

8.1 Types of arbitrage

In practice one distinguishes pure arbitrage (riskless, fleeting, accessible only to the fastest), statistical arbitrage (a probabilistic bet on the convergence of a related-asset spread), and risk arbitrage (event-driven, e.g. mergers, with binary outcomes). Only statistical arbitrage is realistically accessible to a mid-frequency researcher, and it is a bet on a *tendency*, not a certainty.

8.2 Regime detection

Because a strategy may work in only one regime, detecting regime is often essential. Hamilton (1989) introduced the Markov regime-switching model, in which an unobserved state follows a Markov process and the econometrician infers it probabilistically from observed returns—the foundation for hidden-Markov regime detection. In practice, the simplest robust tool is a volatility threshold (for example, scaling exposure down as a volatility index rises); ML classifiers are more flexible but more prone to overfitting. The crowding risk documented by Khandani and Lo (2007) in the August 2007 "Quant Quake"—when many similar market-neutral funds de-leveraged simultaneously and moved prices against their shared positions—shows that regime risk for statistical arbitrage is partly endogenous to the strategy's own popularity.

9. Time Horizons and Instruments

9.1 Horizons

Strategy, horizon, data, and cost are tightly coupled. High-frequency market making (microseconds to seconds) is infrastructure- and data-prohibitive. Mid-frequency statistical arbitrage (hours to a few weeks) runs on free daily data plus optional intraday bars and is the accessible "sweet spot." Trend following (weeks to months) and factor investing (months) are robust but slower to iterate.

9.2 Instruments

Instrument choice determines liquidity, leverage, expiry behaviour, and cost. Equities have no expiry and are flexible, suiting reversion, momentum, and factor strategies. Futures embed leverage (via margin) and must be rolled, suiting time-series momentum. Options are fundamentally instruments of volatility rather than direction—their value is driven by expected movement, so they are bought when implied volatility is too low and sold when it is too high—and require managing the Greeks (delta, gamma, theta, vega). Commodities carry storage costs and seasonality and exhibit persistent trends. The relevant reference for instrument mechanics is Hull's *Options, Futures, and Other Derivatives*.

10. Evaluation Metrics

The decisive theme. A Sharpe ratio alone misleads because the best of many tried strategies is inflated by luck (multiple testing). Bailey and López de Prado (2014) formalize the **Deflated Sharpe Ratio**, which corrects an observed Sharpe for the number of trials, the sample length, and the skewness and kurtosis of returns, and reports the probability that the true Sharpe is positive. Evaluation therefore reports a panel—net (after-cost) Sharpe and Sortino, maximum drawdown and duration, win rate together with the average win/loss ratio, and turnover—validated by *walk-forward analysis* (train on the past, test on the future, roll forward) and a regime breakdown. The methodological requirement is honesty: every variation tried must be recorded, or the deflation is meaningless, and the final out-of-sample read must be taken once and not re-optimized.

11. Comparative Analysis

11.1 By regime

Strategy	**Calm / trending**	**Crisis / high-volatility**
Mean reversion / stat-arb	Works (divergences revert)	Fails (divergences become real; correlations $\to 1$)
Momentum / trend	Works (trends persist)	Crashes at turning points
Factor (value/quality)	Works over months	Regime-dependent; can invert

11.2 By timeframe, instrument, and viability

Strategy	Horizon	Best instrument	Deployment viability (this project)
PCA-residual stat-arb	Hours–weeks	Equities	**High**: free baseline data, fits horizon and skills; cost-constrained
Pairs trading	Days–weeks	Equities	High: simplest baseline; less scalable
Time-series momentum	Weeks–months	Futures	High: robust, few parameters; slower than target
Cross-sectional factors	Monthly	Equities	Medium: clean fundamentals harder to source; slow iteration
Order-book imbalance	Seconds–minutes	Equities/futures (LOB)	Low: LOB data and latency prohibitive

12. Table of Papers and Key Findings

Reference	Theme	Key finding used in this review
Asness, Moskowitz & Pedersen (2013), *J. Finance* 68(3)	Classical / factors	Value and momentum premia are pervasive across eight markets; value–momentum correlation ≈ -0.5 to -0.6.
Gatev, Goetzmann & Rouwenhorst (2006), *RFS* 19(3)	Stat-arb / pairs	Pairs trading earns up to $\sim$11% annualized excess return; profit from temporary mis-pricing of substitutes.
Avellaneda & Lee (2010), *Quant. Finance* 10(7)	Stat-arb (selected)	PCA-residual mean reversion via OU process; PCA Sharpe ≈ 0.9 (2003–07), modest and decaying.
Fama & French (2015), *JFE* 116(1)	Stock selection	Five-factor model (Mkt, SMB, HML, RMW, CMA); profitability/investment add value, HML partly redundant.
Cartea, Jaimungal & Penalva (2015), CUP	Microstructure	Mathematical treatment of order-book signals, optimal execution, and market making.
Glosten & Milgrom (1985), *JFE* 14(1)	Market structure	Bid–ask spread arises from adverse selection; set by competition, widens with informed flow.
Kyle (1985), *Econometrica* 53(6)	Market structure	Price impact of informed trading; the "Kyle's lambda" notion of impact.
López de Prado (2018), Wiley (AFML)	ML / DL	Why ML fails on markets and the fixes: triple-barrier labels, purged CV, meta-labelling, fractional differentiation.
Hamilton (1989), *Econometrica* 57(2)	Regime	Markov regime-switching model; infer hidden state probabilistically from observed returns.
Khandani & Lo (2007), *J. Investment Mgmt* 5(4)	Regime / risk	The August 2007 "Quant Quake": crowded de-leveraging causes simultaneous losses in similar strategies.
Lo (2004), *JPM* 30(5)	Market setting	Adaptive Markets Hypothesis: efficiency is evolutionary; opportunities migrate and decay.
Bailey & López de Prado (2014), *JPM* 40(5)	Evaluation	Deflated Sharpe Ratio corrects for trials, sample length, and non-normality.
Hull, *Options, Futures, and Other Derivatives*	Instruments	Standard reference for derivative mechanics, leverage, expiry, and the Greeks.

13. Conclusion and Recommended Direction

The survey supports a clear synthesis. Predictability is conditional and perishable (Lo); each strategy family is tied to a regime, horizon, and instrument; microstructure sets a hard cost filter on what is tradable (Glosten–Milgrom, Kyle); ML helps only narrowly and only on top of a working baseline (López de Prado); and honest, deflated, out-of-sample evaluation is what distinguishes real edge from luck (Bailey–López de Prado). None of these factors acts alone; a viable strategy is a consistent joint choice across all of them.

13.1 Recommended directions

The most promising direction for this project is **PCA-residual statistical arbitrage**: a market-neutral, mid-frequency mean-reversion strategy on liquid Indian equities, holding from intraday to a few weeks. It fits the accessible data (free daily data for the factor structure), the target horizon, and an optimization-oriented skill set, and it offers a clean build path (classical baseline, then ML for sizing and exits, then an intraday extension). Its edge is known to be modest and cost-sensitive, which makes it an honest hypothesis to test rather than a guaranteed system—fully consistent with the brief's emphasis that negative results, properly analysed, are valuable.

Robust alternative: time-series momentum on Indian index futures with a volatility regime filter—fewer parameters, lower overfitting risk, and strong historical robustness, at the cost of a slower horizon than the primary target.

Bottom line. Choose a strategy as a joint bet on regime, horizon, instrument, cost, and method—then test it honestly. The recommended direction (PCA-residual stat-arb) is selected not because it is guaranteed to profit, but because it is a well-posed, testable, cost-aware hypothesis that fits this project's constraints.

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